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PROFILE

Second-year CS (AI & ML) student who builds production-shaped ML systems and works primarily through AI agents. Currently building **UnlearnShield** — provable data deletion for fraud models, with signed deletion certificates, a Postgres-backed job queue and 241 tests, validated against 2M real payment transactions. Two merged fixes to **Scrapling**, one hardening its MCP server against unauthenticated access. First-named inventor on a filed provisional patent for real-time cognitive state estimation, and author of an 87,000-repository semantic search system whose ranking model is an LLM distilled into a from-scratch gradient-boosted regressor.

SELECTED PROJECTS

UnlearnShield — provable data deletion for fraud models 2026, ongoing
Python, PyTorch, XGBoost, PostgreSQL, FastAPI, Docker · github.com/yamantaka-singh/unlearn-shield

Exact unlearning, not approximate. SISA architecture — data sharded, isolated and sliced — so a deletion request retrains one shard from a rollback point instead of the whole model. On the gradient-boosted engine rollback is tree truncation, verified bit-identical to training that many rounds from scratch; that verification is what separates “the subject's data is gone” from a claim nobody can check.

Deletion certificates that name nobody. Every erasure emits an Ed25519-signed manifest carrying a sparse-Merkle absence proof over the retained set. Chose sparse over sorted Merkle after finding the sorted construction discloses two other subjects' identifiers in every certificate it issues.

Held-out validation on real data, and the metric changed. Ran 2M real PaySim transactions on a temporal split. Stopped reporting ROC-AUC after showing it reads 0.9988 where average precision reads 0.8347 at 0.067% fraud prevalence. Real data also exposed two bugs synthetic data could not: an empty-shard crash, and a hardcoded time horizon that silently disabled hot-shard concentration entirely.

Profiled before optimising, and retracted a wrong number. Erasure took ~90s, of which the retraining SISA exists to shorten was 1%. An earlier breakdown derived by subtraction hid a duplicate Merkle traversal; measuring each phase directly found it. Building the tree once per rebuild cut erasure to 51s, and persisting it across erasures took repeat deletions on a hot shard from 42s to ~0.2s — behind an O(n) fingerprint check that falls back to a full rebuild rather than ever trust a stale tree.

Operational shape. PostgreSQL job queue (FOR UPDATE SKIP LOCKED, leases, crash recovery), content-addressed checkpoints, HMAC-sampled reproducibility spot-checks an operator cannot steer, read-only dashboard role, determinism harness pinning threads and seeds. 241 tests, 12 architecture decision records. Found and closed a cross-tenant idempotency-key leak while adversarially reviewing my own gateway.

GitGlobe — semantic map of the open-source ecosystem 2026
Python, PostgreSQL, NumPy, TypeScript, WebGL · gitglobe-yd-mj.vercel.app

LLM → model distillation with a popularity blindfold. An LLM teacher (NVIDIA Nemotron 550B via NIM) rates a stratified sample against a six-dimension quality rubric; a gradient-boosted tree regressor written from scratch in NumPy distils those judgements to all 95,384 rows. Star and fork counts are stripped from both the teacher's prompt and the student's features, enforced by tests — without that, the model simply relearns popularity.

Refused to ship a model that had not earned it. A dimension is stored only if its held-out RMSE beats the mean-predictor by more than sampling noise (baseline/√2n). Two of six dimensions failed that bar across repeated runs and are not stored; four passed at R^2 0.21–0.44.

Systems engineering. Custom binary tile format (12 bytes/point, 3 LOD bands, 3.2 MB for 87,227 nodes); CSR graph of 234,640 entries with PageRank; UMAP projection to 3-D and spherical k-means clustering; WebGL renderer with GPU picking. 446 tests across 19 suites, with NASA Power-of-10 rules enforced by a test rather than by convention.

OPEN SOURCE

Scrapling — adaptive web-scraping framework · two merged fixes, shipped in v0.4.15 (August 2026)

MCP server hardening (#414). The HTTP transport served every tool unauthenticated and bound to 0.0.0.0, so anyone who could reach the port could fetch arbitrary URLs through the host. Made --http refuse to start without a token or an explicit --no-auth, and moved the default bind to localhost — separating reachability from authorisation instead of logging a warning and continuing. Shipped as a documented breaking change, with tests and agent-skill docs updated.

Parser correctness (#417). find/find_all silently dropped multi-class elements when class_ was blank, and CSS string values went unescaped.

HOW I WORK WITH AI

Every project above was built with AI agents as the primary implementation tool. The engineering discipline is in what surrounds them.

Verification over generation. Agent output is assumed wrong until measured. Real defects this caught: a query that silently discarded 84,434 backfilled values because an absent dictionary key became a neutral default; a scoring component saturated across an entire corpus and ordering nothing; an apparent accuracy collapse in UnlearnShield that root-caused to a module-binding bug in my own validation script — recorded as a retraction rather than quietly corrected.

Tests as the specification. Each guard is itself tested for its ability to fail; a check that cannot fail is decoration. One UnlearnShield test exists purely to prove it fails when a specific module binding is left unpatched, reproducing a real incident on demand.

Spec-first, agent-executed. Architecture decision records and written implementation plans precede code; agents execute against them. 12 ADRs across UnlearnShield alone.

Tooling, not just usage. Primary environment is Claude Code. Authored reusable agent skills and plugins so recurring workflows stay repeatable, and contributed a security fix to an MCP server's transport layer.

RESEARCH & PATENTS

Provisional Patent (First Inventor) — Real-Time Cognitive State Estimation & Environmental Modulation

Indian Patent Office · App. No. IN202611015895 · Filed 12 February 2026 · Project S.A.G.E.

Architecture. A hardware-agnostic design building a "cognitive digital twin" of an operator through multi-modal sensor fusion and closed-loop feedback, keyed to quantifiable signal variances (e.g. keystroke flight-time deviation) rather than abstract psychometric traits, with consent-bounded fail-safe intervention protocols. Led a three-person interdisciplinary team through specification and filing.

Causal design behind it. Authored a multicentre randomised controlled protocol (N = 240, 1:1 stratified randomisation, assessor-blinded) and its statistical analysis plan: power justification, ITT/PP populations, multiplicity control, pre-specified subgroups, interim analysis — the machinery for showing an intervention causes a behavioural change rather than correlating with one.

Lead Author — "Cognitive Architecture and Stochastic Algorithmic Reinforcement"

A Systems Model of Adolescent Vulnerability · Singh, Dabas & Kanan, 2026 · SSRN

Modelled how variable-ratio reinforcement schedules in recommendation engines interact with delayed prefrontal maturation to shift users toward fast associative processing — a mechanistic account of engagement optimisation and the long-term user value it trades against.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript/JavaScript, SQL, C, GLSL, Bash, HTML/CSS

ML & Modelling: PyTorch (functional_call/vmap for batched shard ensembles, deterministic training), XGBoost, NumPy, gradient boosting implemented from scratch, knowledge distillation, held-out and temporal-split evaluation, precision-recall metrics at low prevalence, UMAP, k-means/HDBSCAN, PageRank, scikit-learn

Agentic AI: Claude Code as primary development environment, MCP servers (contributed a transport security fix), agent skill and plugin authoring, structured-output prompting with schema validation, rubric design, leakage guards, teacher–student distillation

MLOps & Infrastructure: PostgreSQL (schema design, migrations, job queues with SKIP LOCKED leasing, multi-million-row joins), FastAPI, Docker, CI/CD, content-addressed artifact storage, model versioning and promotion, reproducibility and determinism harnesses, Google Cloud (Vertex AI, BigQuery), Git/GitHub, Linux

Applied Cryptography: Ed25519 signing, Merkle and sparse-Merkle absence proofs, HMAC-based sampling

Quantitative Methods: Randomised controlled trial design, statistical analysis planning, power analysis, stratified sampling, linear algebra, probability, graph theory

Currently building depth in: transformer internals; causal inference estimators (difference-in-differences, propensity scoring, uplift modelling); AWS (S3, Athena, EC2)

EDUCATION

Chandigarh University, Mohali — B.E. (Hons.) Computer Science (AI & ML), Minor in Geoinformatics 2025 – 2029

50% merit scholarship for academic performance; honours track in machine learning and deep learning architectures.

Siddharth Public School — Class XII 92.6%, Class X 94.2%; JEE Main 2024: 93.95 percentile.

LEADERSHIP

School Captain — primary liaison between management and a 3,000+ student body; led event logistics and cross-functional teams. Class Representative for seven consecutive years. 1st Place, District Essay Competition (Siddharthnagar).